Memory

How information gets in, how it's stored, why it fades, and why confident memories are so often wrong — with what that means for training, advertising and settling disputes.

Isagi · November 2026 · Spine: Ciccarelli & White, *Psychology* (4th ed.), Ch. 6 — rewritten, updated and applied

A company spends a day training its sales force on a new product range. A week later, most of the reps remember the lunch better than the features. An ad runs for a month; ask viewers a week after it stops and most can't name the brand. A dealer insists he was promised a 5% discount; the rep insists he said 3%. Both are sincere. All three are memory problems, and memory research has unusually practical answers to them.

Memory is the system that encodes, stores and retrieves information. Each step can fail. **Encoding** is getting information into a form the brain can store. **Storage** is holding it over time. **Retrieval** is getting it back out when needed.

1 The three-stage model

The most useful map is the **information-processing model** (Atkinson and Shiffrin, 1968). Information passes through three stores, each with its own capacity and duration:

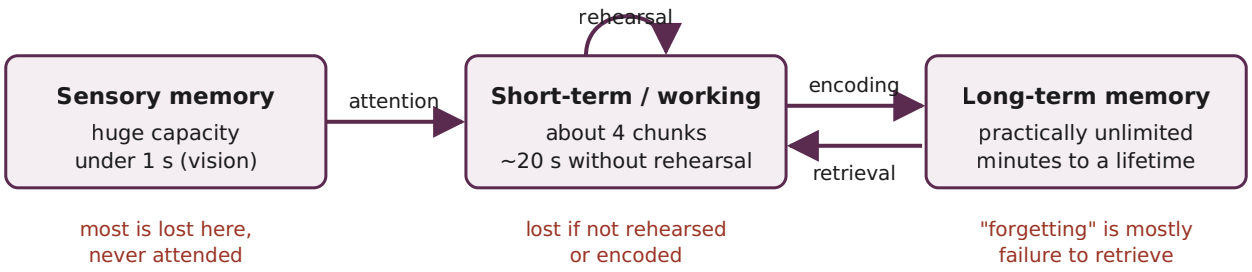


Figure 1. The three-stage model. Attention moves information from sensory memory into working memory; encoding moves it into long-term memory; retrieval brings it back.

Sensory memory

Every sense holds a brief, detailed copy of its input. In 1960 George Sperling flashed a grid of 12 letters for a twentieth of a second. People could report only about four — but if a tone told them which row to report *after* the grid vanished, they could report almost any row. The whole image was there for a fraction of a second, then faded. Visual (**iconic**) memory lasts well under a second; auditory (**echoic**) memory a few seconds, which is why you can "replay" what someone just said. Only what you attend to moves on.

Short-term and working memory

Short-term memory (STM) holds what you are thinking about right now. Two limits define it. **Duration**: without rehearsal, most of it is gone within about 20 seconds. **Capacity**: George Miller's famous 1956 estimate was "seven, plus or minus two" items; more recent work puts the true limit nearer four meaningful chunks.

Chunking — grouping items into meaningful units — is how we beat the limit. Alan Baddeley and Graham Hitch (1974) refined STM into **working memory**: an active workspace with a central controller (the **central executive**), a verbal store you hear in your head (the **phonological loop**), and a visual one (the **visuospatial sketchpad**). A dealer reading out a long order while you also check stock on screen is loading both, and the central executive is the bottleneck.

WORKED EXAMPLE 1 · CHUNKING

A rep must remember a dealer code: 4 0 7 2 1 9 8 3 5 6 1.

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As single digits:   11 items → well over capacity
As chunks:          407 · 219 · 835 · 61  → 4 chunks
With meaning:       "Kochi-407", "2019", "835", "61"
                    → easier still
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Same information, a third the load. This is why phone numbers, PINs and SKU codes are printed in groups, and why a pitch with "three reasons" is remembered and one with eleven isn't. In product training, organise features into a few named groups before presenting the details.

Two kinds of rehearsal keep information alive. **Maintenance rehearsal** — repeating it over and over — holds it in STM but transfers it poorly to long-term memory. **Elaborative rehearsal** — linking it to what you already know, asking what it means — encodes it much more durably. Craik and Lockhart's **levels-of-processing** idea (1972) makes the same point: the more deeply you think about the meaning of something, the better you remember it.

2 Long-term memory

Long-term memory (LTM) has no practical capacity limit. It holds two broad kinds of knowledge:

- **Procedural (implicit) memory** — skills and habits: driving, typing, the routine of a dealer visit. Often hard to put into words, and resistant to forgetting.
- **Declarative (explicit) memory** — facts you can state. It splits into **semantic** memory (general knowledge: the price list, the capital of Kerala) and **episodic** memory (personal events: what happened at last month's dealer meet).

LTM is organised by meaning. One model pictures it as a **semantic network** of linked ideas: think "premium emulsion" and related nodes — price, brand, a particular customer — become easier to reach. This is why a strong brand association pays off at the shelf: the cue triggers the whole network.

3 Getting it back out

Retrieval cues are stimuli that help you access a memory. The **encoding specificity** principle: memory is best when the conditions at retrieval match those at encoding. In a classic 1975 study, divers who learned word lists underwater recalled them better underwater, and those who learned on land recalled better on land. Train reps in conditions that resemble the field — role-plays with real dealer objections, not slides in a hotel ballroom.

Recognition (picking the right answer from options) is much easier than **recall** (producing it from nothing). That's why customers who can't name your brand will still pick it out on a shelf — and why brand-recall surveys understate what a brand has achieved.

The **serial position effect**: in a list, people remember the first items (**primacy**, because they got more rehearsal) and the last ones (**recency**, still in working memory) better than the middle. If you're one of six agencies pitching, avoid the middle slots. In a presentation, put your key point first and repeat it last.

The testing effect

Retrieving information strengthens it more than re-reading it. In a 2006 study by Henry Roediger and Jeffrey Karpicke, students who studied a passage once and then took recall tests forgot far less a week later than students who re-read it several times — even though the re-readers felt more confident. Quizzes after training aren't just assessment; they are the training.

4 Memory is reconstructed, not replayed

Memory is not a recording. Each time you remember, you rebuild the event from fragments, filling gaps with what is likely, what you expect and what you've heard since.

Elizabeth Loftus and John Palmer (1974) showed people a film of a car crash, then asked how fast the cars were going when they "smashed into" or "contacted" each other. The "smashed" group estimated about 41 mph on average; the "contacted" group about 32 mph. A week later, people in the "smashed" group were about twice as likely to say they had seen broken glass. There was none. One word changed the memory itself. This is the **misinformation effect**.

Later work went further: by repeatedly suggesting a childhood event, such as being lost in a shopping mall, researchers led a minority of participants (about a quarter in one early study) to "remember" an event that never happened. Even vivid **flashbulb memories** of dramatic events fade and drift like ordinary memories; what stays high is the person's confidence.

For a manager the lesson is practical. In a dispute over what was promised, both sides can be honest and wrong, and confidence proves nothing. Write it down: confirm terms on WhatsApp or email the same day. When investigating an incident, ask open questions ("What happened next?") rather than leading ones ("Did he shout at you?").

5 Forgetting

In the 1880s Hermann Ebbinghaus memorised lists of nonsense syllables and measured how much he retained over time. The **forgetting curve** drops steeply at first, then levels off:

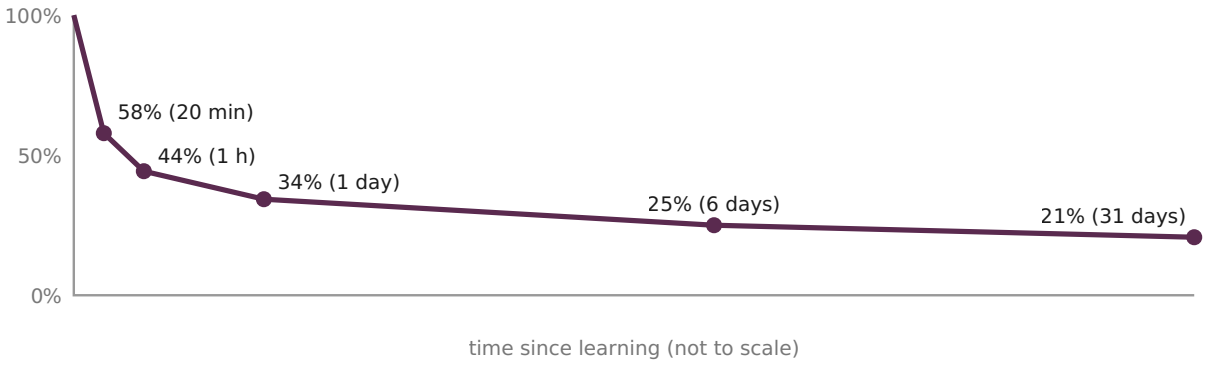


Figure 2. Ebbinghaus's forgetting curve, measured as "savings" (how much faster he relearned a list). Most loss happens within the first day. Meaningful material is forgotten more slowly than nonsense syllables, but the shape is the same.

Why do we forget?

- **Encoding failure** — it never got in. In a classic 1979 study, most Americans couldn't pick the correct design of a penny from a set of near-copies, despite handling thousands. They never needed the detail, so it was never encoded.
- **Decay** — unused memory traces may fade over time, though this is hard to prove.
- **Interference** — other memories get in the way. **Proactive** interference: old learning disrupts new (last year's price list keeps surfacing). **Retroactive** interference: new learning disrupts old (after learning the new scheme, you can't recall the old one's details).
- **Retrieval failure** — the memory is there but the cue is wrong: the tip-of-the-tongue feeling.

WORKED EXAMPLE 2 · A TRAINING PLAN THAT SURVIVES THE WEEK

On Monday, reps learn 20 key facts about a new range. Using Ebbinghaus's shape as a rough guide, by the next Monday they might retain about a quarter — 5 facts — if nothing else happens.

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Session  (Mon):  learn all 20, end with a recall quiz
Review 1 (Tue):  10-minute quiz on WhatsApp
Review 2 (Thu):  role-play using the facts
Review 3 (Mon+): quiz on the 5 most-missed facts
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This uses three robust findings at once: the **testing effect** (quizzes, not re-reading), the **spacing effect** (reviews spread out beat one long session), and **encoding specificity** (role-play resembles the field). Total extra time: under an hour. Expected result: most of the 20, not five.

6 The biology of memory

New long-term memories are stabilised over hours to years in a process called **consolidation**, which relies heavily on the **hippocampus** (Part 1) and on sleep (Part 3). The most studied patient in psychology, Henry Molaison ("H.M."), had parts of both hippocampi removed in 1953 to control epilepsy. Afterwards he could not form new declarative memories — **anterograde amnesia** — though his older memories and his ability to hold a conversation for a few minutes survived. Yet he improved at a mirror-drawing task over days without remembering ever having done it. Procedural and declarative memory depend on different systems. **Retrograde amnesia**, the loss of memories from before an injury, is the reverse. **Alzheimer's disease** typically attacks the hippocampus early, which is why recent memories go first.

7 Held up?

Claim	Verdict	Why
People learn best in their preferred "learning style" (visual, auditory, kinaesthetic)	**No**	A 2008 review found almost no evidence that matching teaching to a person's preferred style improves learning. Match the method to the material instead.
Memory works like a video recording	**No**	It's reconstructed each time; the misinformation effect is one of the best-replicated findings in the field.
Testing and spacing improve retention	**Robust**	Replicated across hundreds of studies, ages and materials.
Brain-training games make you smarter	**Little evidence**	A 2016 review found people get better at the trained games but gains rarely transfer to everyday thinking.
Listening to Mozart raises IQ	**No**	The original small effect was brief, came from arousal and mood, and failed to replicate reliably.

Summary

- Memory involves **encoding, storage** and **retrieval**, and can fail at each step.
- **Sensory memory** holds a full copy for under a second; only what you attend to moves on.
- **Working memory** holds about four chunks for about 20 seconds without rehearsal. **Chunking** beats the limit; **elaborative rehearsal** encodes far better than repetition.
- **Long-term memory** is effectively unlimited: procedural (skills) and declarative (semantic facts, episodic events), organised by meaning.
- Retrieval is best when conditions match learning (**encoding specificity**). Recognition beats recall. First and last items are remembered best. Testing yourself beats re-reading.
- Memory is **reconstructed**: one word in a question changed speed estimates and created memories of broken glass. Confidence isn't accuracy — write agreements down.
- Forgetting is steep at first, then levels off. Causes: encoding failure, decay, interference, retrieval failure. Spaced quizzes after training preserve most of what was taught.
- The **hippocampus** consolidates new declarative memories; H.M. lost that ability but could still learn skills.
- "Learning styles", the Mozart effect and brain-training games don't hold up.

Practice problems

P1. Chunk this order number for a rep to read back on the phone: 2 6 1 1 0 4 4 7 8 1 2 0. How many chunks did you use?

P2. A brand runs a recall survey ("Which paint brands can you name?") and a recognition survey (shows logos). It scores 18% on recall and 64% on recognition. Is the brand failing? Explain.

P3. You're the fourth of seven presenters at a client pitch. What does the serial position effect suggest, and what can you do about it?

P4. Investigating a complaint, a manager asks a retailer, "Did the rep rudely refuse to replace the damaged tins?" What's wrong with the question? Rewrite it.

P5. After switching to a new CRM, reps keep entering orders the old way. Which type of interference is this?

P6. An L&D head proposes splitting reps into "visual" and "auditory" learners for training. Using the evidence, what would you recommend instead?

Glossary

Encoding / storage / retrieval

Getting information in / keeping it / getting it out.

Sensory memory

A very brief, detailed copy of sensory input.

Working memory

The active workspace for current information, about four chunks.

Chunking

Grouping items into meaningful units to reduce memory load.

Maintenance / elaborative rehearsal

Simple repetition / linking new information to what you know.

Levels of processing

Deeper, meaning-based processing produces stronger memories.

Procedural / declarative memory

Skills and habits / facts and events you can state.

Semantic / episodic memory

General knowledge / personal experiences.

Encoding specificity

Memory is best when retrieval conditions match encoding conditions.

Recall / recognition

Producing information unaided / identifying it from options.

Serial position effect

Better memory for the first (primacy) and last (recency) items in a list.

Testing effect

Retrieving information strengthens memory more than re-studying.

Spacing effect

Study spread over time beats the same amount crammed together.

Misinformation effect

Later misleading information distorts the memory of an event.

Proactive / retroactive interference

Old learning disrupts new / new learning disrupts old.

Consolidation

The process that stabilises new long-term memories.

Anterograde / retrograde amnesia

Inability to form new memories / loss of earlier ones.

Answer key

P1. For example 2611 · 0447 · 8120 (three chunks) or 261 · 104 · 478 · 120 (four). Either is within capacity; twelve single digits are not.

P2. Not necessarily. Recall is much harder than recognition, so recall scores always run lower. 64% recognition means most customers will spot it on a shelf. Whether 18% recall is a problem depends on category norms and how often customers choose without the shelf in front of them.

P3. Middle positions are remembered least. Ask for the first or last slot if you can; if not, make your opening distinctive, keep the core message to a few chunks, and end with a one-line restatement.

P4. It is a leading question: it suggests both rudeness and refusal, which can change the retailer's memory. Better: "Tell me what happened when you raised the damaged tins with the rep."

P5. Proactive interference: old learning disrupts the new.

P6. Drop the style grouping; the evidence for matching doesn't hold up. Use methods that work for everyone: chunk the content, use role-plays that match the field, quiz instead of re-reading, and space reviews over the following week.

Sources: chapter structure follows Ciccarelli & White, *Psychology*, 4th ed. (Pearson), Ch. 6; text and figures are original. Sperling, 1960. Miller, 1956; Cowan, *Behavioral and Brain Sciences*, 2001. Baddeley & Hitch, 1974. Godden & Baddeley, 1975. Loftus & Palmer, 1974. Roediger & Karpicke, *Psychological Science*, 2006. Learning styles: Pashler et al., *Psychological Science in the Public Interest*, 2008. Brain training: Simons et al., *Psychological Science in the Public Interest*, 2016. Ebbinghaus, 1885. Worked-example figures are illustrative.